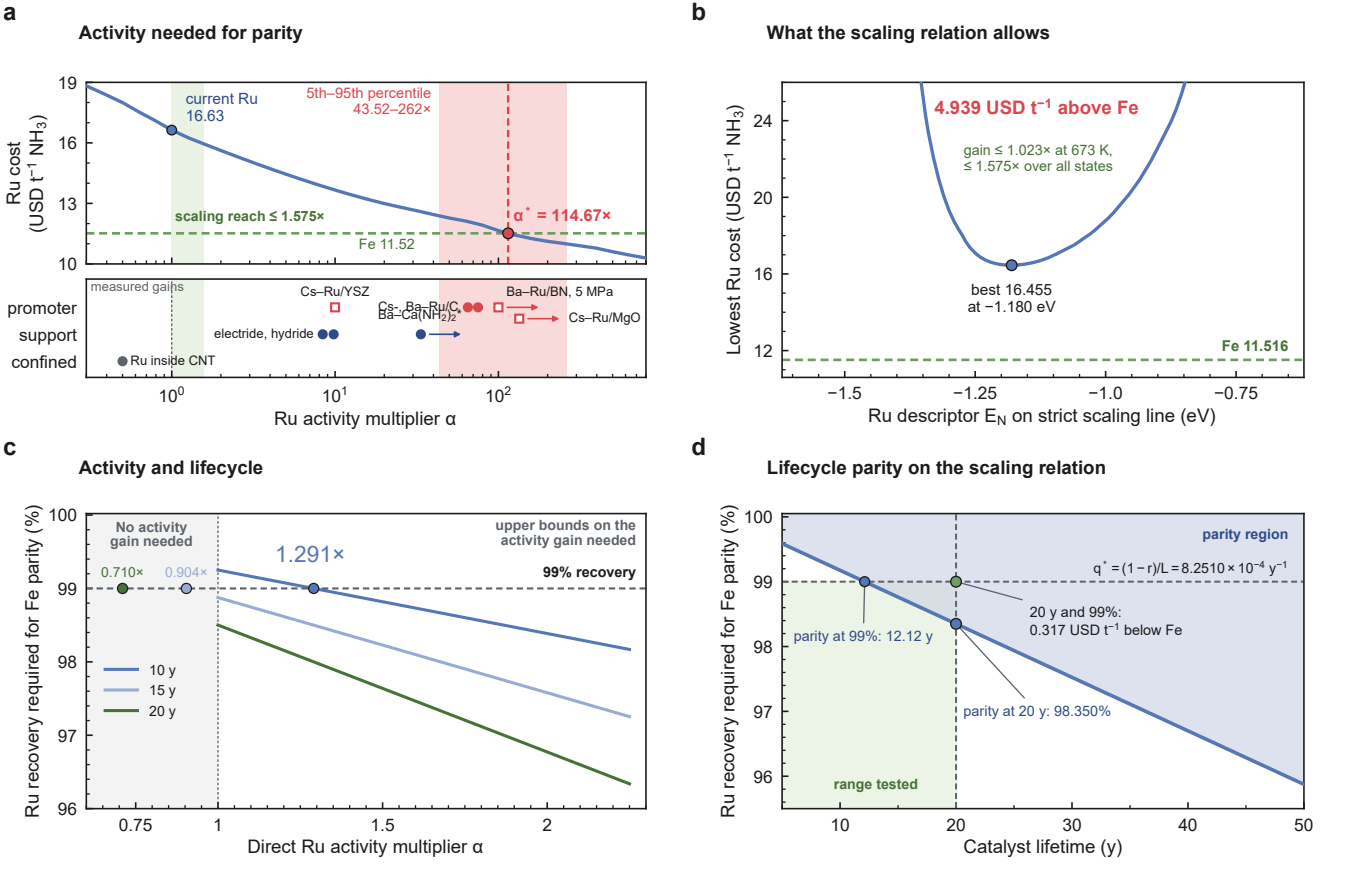

a
Activity needed for parity
Ru cost (USD t⁻¹ NH3)
19
16
13
10
current Ru 16.63
5th–95th percentile 43.52–262×
scaling reach ≤ 1.575×
α* = 114.67×
Fe 11.52
measured gains
promoter
support
confined
Cs–Ru/YSZ
Cs–Ba–Ru/C
Ba–Ca(NH2)2
Ba–Ru/BN, 5 MPa
Cs–Ru/MgO
electride, hydride
Ru inside CNT
10^0
10^1
10^2
Ru activity multiplier α
b
What the scaling relation allows
Lowest Ru cost (USD t⁻¹ NH3)
24
20
16
12
4.939 USD t⁻¹ above Fe
gain ≤ 1.023× at 673 K,
≤ 1.575× over all states
best 16.455
at −1.180 eV
Fe 11.516
−1.5
−1.25
−1.0
−0.75
Ru descriptor EN on strict scaling line (eV)
c
Activity and lifecycle
Ru recovery required for Fe parity (%)
100
99
98
97
96
No activity gain needed
upper bounds on the activity gain needed
0.710×
0.904×
1.291×
99% recovery
10 y
15 y
20 y
0.75
1
1.5
2
Direct Ru activity multiplier α
d
Lifecycle parity on the scaling relation
Ru recovery required for Fe parity (%)
100
99
98
97
96
parity region
q* = (1 − r)/L = 8.2510 × 10⁻⁴ y⁻¹
20 y and 99%:
0.317 USD t⁻¹ below Fe
parity at 99%: 12.12 y
parity at 20 y: 98.350%
range tested
10
20
30
40
50
Catalyst lifetime (y)